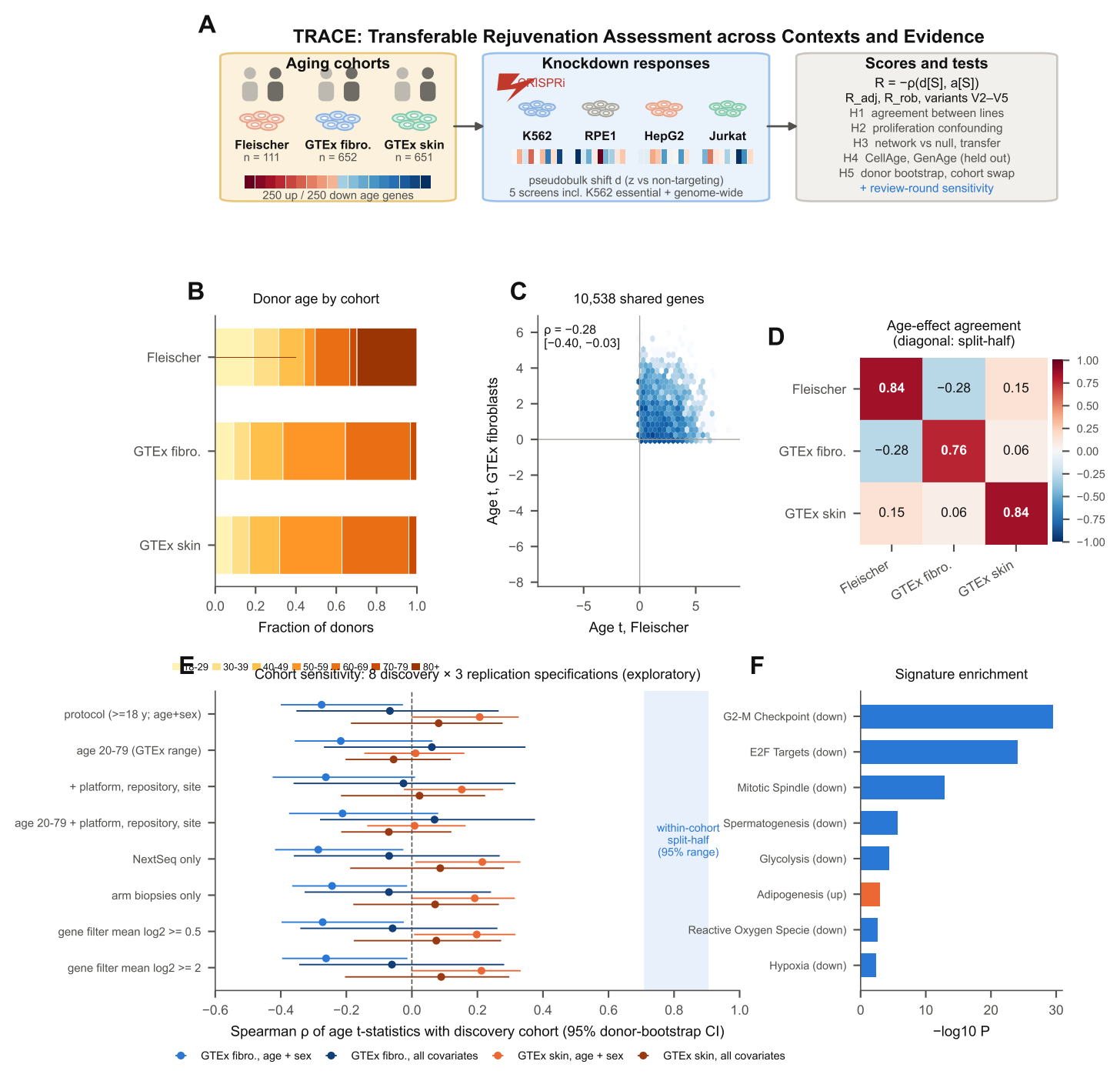

A
TRACE: Transferable Rejuvenation Assessment across Contexts and Evidence
Aging cohorts
Fleischer
n = 111
GTEx fibro.
n = 652
GTEx skin
n = 651
250 up / 250 down age genes
Knockdown responses
CRISPRi
K562
RPE1
HepG2
Jurkat
pseudobulk shift d (z vs non-targeting)
5 screens incl. K562 essential + genome-wide
Scores and tests
R = −ρ(d[S], a[S])
R_adj, R_rob, variants V2–V5
H1 agreement between lines
H2 proliferation confounding
H3 network vs null, transfer
H4 CellAge, GenAge (held out)
H5 donor bootstrap, cohort swap
+ review-round sensitivity
B
Donor age by cohort
Fleischer
GTEx fibro.
GTEx skin
0.0
0.2
0.4
0.6
0.8
1.0
Fraction of donors
C
10,538 shared genes
ρ = −0.28
[−0.40, −0.03]
Age t, GTEx fibroblasts
Age t, Fleischer
D
Age-effect agreement
(diagonal: split-half)
Fleischer
GTEx fibro.
GTEx skin
0.84
−0.28
0.15
−0.28
0.76
0.06
0.15
0.06
0.84
1.00
0.75
0.50
0.25
0.00
−0.25
−0.50
−0.75
−1.00
E
20-29
30-39
40-49
50-59
60-69
70-79
80+
Cohort sensitivity: 8 discovery × 3 replication specifications (exploratory)
protocol (>=18 y; age+sex)
age 20-79 (GTEx range)
+ platform, repository, site
age 20-79 + platform, repository, site
NextSeq only
arm biopsies only
gene filter mean log2 >= 0.5
gene filter mean log2 >= 2
within-cohort
split-half
(95% range)
−0.6
−0.4
−0.2
0.0
0.2
0.4
0.6
0.8
1.0
Spearman ρ of age t-statistics with discovery cohort (95% donor-bootstrap CI)
GTEx fibro., age + sex
GTEx fibro., all covariates
GTEx skin, age + sex
GTEx skin, all covariates
F
Signature enrichment
G2-M Checkpoint (down)
E2F Targets (down)
Mitotic Spindle (down)
Spermatogenesis (down)
Glycolysis (down)
Adipogenesis (up)
Reactive Oxygen Specie (down)
Hypoxia (down)
0
10
20
30
−log10 P